LATIHAN KARNAUGH MAPS

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PART I

5-1. The K-map of a 3-variable function F is given below. What is the sum-of-minterms expression of F?

$a \backslash bc$	00	01	11	10
0	**1**	0	0	**1**
1	0	**1**	0	0

5-2. Draw the K-map for this function: $A(x,y,z) = x{\cdot}y + y{\cdot}z' + x'{\cdot}y'{\cdot}z$.

5-3. Identify the prime implicants and essential prime implicants for the K-maps below.

a.

$a \backslash bc$	00	01	11	10
0	**1**	**1**	0	**1**
1	0	**1**	0	0

b.

$CD \backslash AB$	00	01	11	10
00	**1**	**1**	0	**1**
01	0	0	**1**	**1**
11	**1**	**1**	0	**1**
10	**1**	0	**1**	**1**

5-4. Find the minimal SOP expression for $G(A,B,C,D)$, whose K-map is shown below.

$CD \backslash AB$	00	01	11	10
00	0	**1**	0	0
01	0	**1**	**1**	**1**
11	**1**	**1**	**1**	0
10	0	0	**1**	0

For questions 5-5, 5-6 and 5-7, refer to the following K-map of a function $W(A,B,C,D)$.

AB \ CD	00	01	11	10
00	**1**	**1**	0	**1**
01	0	**1**	**1**	0
11	0	**1**	**1**	**1**
10	0	**1**	0	0

5-5. Which of the following is not a prime implicant?

a. $A'\cdot B'\cdot C'$ b. $C'\cdot D$ c. $B\cdot D$ d. $A\cdot B\cdot C$ e. $A\cdot B\cdot D$

5-6 Which of the following is not an essential prime implicant?

a. $A'\cdot B'\cdot D'$ b. $C'\cdot D$ c. $B\cdot D$ d. $A\cdot B\cdot C$ e. $A'\cdot B'\cdot C'$

5-7 What is the minimal SOP expression for W' (the complement of W)?

a. $A\cdot B'\cdot C + A'\cdot B\cdot D' + A\cdot C'\cdot D' + B'\cdot C\cdot D$
b. $A\cdot B\cdot C' + A\cdot B'\cdot D' + A'\cdot C\cdot D' + B\cdot C'\cdot D$
c. $A\cdot B'\cdot C + A'\cdot B\cdot D' + A\cdot C'\cdot D + B'\cdot C\cdot D'$
d. $A\cdot B\cdot C + A\cdot B\cdot D' + A\cdot C'\cdot D' + B\cdot C'\cdot D$
e. $A\cdot B\cdot C + A'\cdot B\cdot D' + A\cdot C'\cdot D' + B\cdot C\cdot D'$

For questions 5-8, 5-9 and 5-10, refer to the following K-map of a function $Y(A,B,C,D)$.

AB \ CD	00	01	11	10
00	0	0	X	0
01	**1**	0	X	X
11	X	X	**1**	0
10	0	X	**1**	0

5-8. How many essential prime implicants (EPIs) are there?

a. 0 b. 1 c. 2 d. 3 e. 4

5-9. How many possible minimal SOP expressions for Y are there?

a. 0 b. 1 c. 2 d. 3 e. 4

5-10. Which of the following is a minimal POS expression for Y?

a. $(A + B)\cdot(A + D')\cdot(A' + D)$
b. $(C + D')\cdot(B + D)\cdot(C' + D)$
c. $(A' + D)\cdot(B + D)\cdot(A + D')$
d. $(A' + D)\cdot(B + D)\cdot(C + D')$
e. All of the above.

PART II

5-12.* Draw an appropriate K-map to prove the Consensus Theorem.

5-13.* Use K-map to find the minimal SOP expressions of the following functions.

(a) $f(A,B,C) = \Sigma\, m(0, 2, 3, 5)$
(b) $f(A,B,C) = \Sigma\, m(0, 3, 4, 6, 7)$
(c) $f(A,B,C,D) = \Sigma\, m(1, 2, 7, 12, 15) + d(5, 9, 10, 11, 13)$
(d) $f(A,B,C,D) = \Sigma\, m(0, 2, 5, 15) + d(8, 9, 12, 13)$

5-14.* Find the minimal POS form for the following functions.

(a) $f(A,B,C,D) = \Pi\, M(4, 7, 9, 11, 12) \cdot D(0, 1, 2, 3)$
(b) $f(A,B,C,D) = \Pi\, M(0, 3, 7, 12) \cdot D(2, 10, 11, 14)$

5-15. Design a BCD encrypting circuit that takes in a 4-bit BCD code *ABCD* and encrypts it into another BCD code *EFGH* using the following conversion table:

$0 \rightarrow 1$	$1 \rightarrow 6$	$2 \rightarrow 3$	$3 \rightarrow 7$	$4 \rightarrow 0$
$5 \rightarrow 8$	$6 \rightarrow 2$	$7 \rightarrow 9$	$8 \rightarrow 5$	$9 \rightarrow 4$

(a) Complete the truth table.
(b) Find the minimal SOP expressions for *E*, *F*, *G* and *H*.

5-17.* Given the Boolean expression

$$F(W,X,Y,Z) = \Pi M(1,3,5,8,9,11,12) \cdot d(0,6,10,13)$$

where 'd' represents don't-care.

(a) List out all the prime implicants in the K-map.

(b) List out all the essential prime implicants in the K-map.

(c) What is the simplified SOP expression for *F*?

5-18. Refer to the K-map for a Boolean function *K(A,B,C,D)* as shown below, where **X** represents don't care value.

X	0	0	**X**
X	**X**	**1**	**1**
0	**1**	**X**	0
X	**1**	0	**X**

(a) Express function *K* in big-*M* notation with don't care.

(b) How many prime implicants are there in the K-map? List them out.

(c) How many essential prime implicants are there in the K-map? List them out.

(d) How many possible simplified SOP expressions for *K* are there? List them out.

(e) How many possible simplified POS expressions for *K* are there? List them out.